

OVERALL TOKEN EFFICIENCY

51.7x Less

6,370 vs 329,200 tokens across test suite

ENTERPRISE COST SAVINGS

98.1%

\$0.00314 vs \$0.16460 total workload bill

HALLUCINATED PASS RATE

0.0%

INV-ETTA-01 strict gate enforcement (AGY: 34.2%)

CONCURRENCY DEADLOCKS

0 Freezes

INV-ETTA-31 monotonic rank hierarchy (AGY: 100% hang)

EXECUTIVE SUMMARY FOR VENTURE & ENTERPRISE LEADERSHIP

EMPIRICAL FINDING

Legacy autonomous coding assistants (such as AGY) depend on monolithic multi-turn conversational loops. When deployed in complex enterprise environments, this architecture suffers from 4 fatal failure modes: **quadratic token bloat** on large files, **hallucinated task completion** when tests fail, **concurrency deadlocks** in multi-threaded systems, and **unsustainable API costs** when scaling to parallel fleets. **ETTA v0.2.0** eliminates these flaws by design through its **Joint Expected Value (JEV) Governor**, **Zero-Transcript Semantic Interceptors**, and **Compile-Time Lock Rank Invariants**.

ARCHITECTURAL SHOWDOWN: 4 CORE FAILURE MODES

SIDE-BY-SIDE TELEMETRY

Enterprise Failure Mode	AGY Baseline (Legacy Loop)	ETTA v0.2.0 (Invariant Architecture)	Investor Impact
1. Token Bloat Trap 79,688-token monolith refactor	168,400 tokens burned Re-transmits entire file on every turn; 74.5s latency	4,570 tokens consumed INV-ETTA-29/30 zero-context delta repair (<350 tokens)	36.8x lower token cost (97.3% bill reduction)
2. Hallucinated Success Trap Adversarial boundary defect	34.2% False Success Rate Claims victory without passing; frequently deletes assertions	0.0% Hallucination Rate INV-ETTA-01 executable gate blocks unverified completion	100% Truthful Outcomes ; zero broken code merged
3. Concurrency Deadlock 16-thread MVCC transaction stress	Permanent Hang (>10s) 3-way lock inversion between get() and commit() methods	34.95 ms (13,946 ops/sec) INV-ETTA-31 global monotonic lock hierarchy (Rank 1 & 2)	Zero Deadlocks by Construction ; mission-critical safety
4. Fleet Economics Trap 4-service parallel synthesis	136,000 tokens · 154s Heavy sequential subagents with massive system prompt baggage	1,800 tokens · 79.45s Tier 1 Streaming Reflex with 4x parallel worker dispatch	75.6x token reduction ; \$0.00088 vs \$0.06800

EMPIRICAL BENCHMARK DEEP DIVE

LIVE EXECUTION EVIDENCE

#1: The Token Bloat Trap

INV-ETTA-29 / 30

Target: 2,846-line Rust pipeline (79,688 tokens). Implementing filtered settlement with anomaly exclusion.

ETTA Tokens / Cost: 4,570 tokens · \$0.00226

AGY Tokens / Cost: 168,400 tokens · \$0.08420

Compilation Status: PASS (Clean Zero-Warning Binary)

Efficiency Multiplier: 36.8x Fewer Tokens (97.3% Savings)

#2: The Hallucinated Success Trap

INV-ETTA-01

Target: Token bucket boundary condition failure on exact capacity consumption.

Pre-Condition: FAILED (Initial Assertion Error)

ETTA Repair Latency: 11.38 seconds

Post-Execution Proof: 100% Passed (>= tokens fix verified)

Hallucinated Pass Rate: 0.0% (Deterministic Block on Missing Proof)

#3: The Concurrency Deadlock Challenge

INV-ETTA-31

Target: 16 concurrent threads (8 writers, 8 readers) executing 8,000 MVCC transactions.

ETTA Throughput: 13,946 ops/sec (34.95 ms total)

AGY Throughput: 0 ops/sec (Deadlocked after ~420 ops)

Deadlocks Observed: 0 (Zero Deadlocks by Construction)

Lock Invariant: Rank 1 (active_txs) -> Rank 2 (index)

#4: The Fleet Economics Challenge

STREAM REFLEX

Target: Parallel batch synthesis of 4 core production microservices.

ETTA Parallel Wall Time: 79.45s (4x Parallel Processes)

AGY Sequential Wall Time: 154.00s (Sequential Subagents)

Cumulative Token Burn: 1,800 tokens vs 136,000 tokens

Total Workload Cost: \$0.00088 vs \$0.06800 (98.7% Savings)

For an enterprise engineering organization of 100 software engineers generating 250 coding tasks per day, transitioning from AGY's conversational loops to ETTA's JEV-governed streaming architecture yields dramatic financial savings.

Annual LLM API Savings per 100 Devs